Qualifying Exams Study Guide

Personal study guide made to prepare for PhD qualifying exams at the University of British Columbia. Covers basic analysis (real, complex) and abstract algebra (linear, group theory, ring, field/Galois theory), at the 2nd, perhaps 3rd, year undergraduate level.

Notes by Louis Meunier

1 Analysis

1.1 Real

1.1.1 Fundamentals

Definition 1 (Metric Space): A *metric space* is a set X equipped with a *metric* function $d : X \times X \to \mathbb{R}_{\geq 0}$ that satisfies:

(i) $d(x, y) \geq 0$ and equals 0 iff $y = x$

(ii) $d(x, y) = d(y, x)$

(iii) $d(x, z) \leq d(x, y) + d(y, z)$

Definition 2 (Convergence, Subsequences): A *sequence* $\boldsymbol{x} := \{x_n\} \subset X$ in a metric space X is a function $\boldsymbol{x} : \mathbb{N} \to X$ where we write $x_n = \boldsymbol{x}(n)$. We say $\{x_n\}$ *converges* to some point $x \in X$ if

$$\forall\, \varepsilon > 0, \exists\, N \in \mathbb{N} \text{ s.t. } n \geq N \Rightarrow d(x, x_n) < \varepsilon.$$

We write $x_n \to x$ or $\lim_{n\to\infty} x_n = x$. Equivalently, $x_n \to x$ in (X, d) iff $d(x_n, x) \to 0$ in $(\mathbb{R}, |\cdot|)$.

A *subsequence* of $\boldsymbol{x}$ is a new sequence $\tilde{\boldsymbol{x}}$ which can be written as $\tilde{\boldsymbol{x}} = \boldsymbol{x} \circ n$ where $n : \mathbb{N} \to \mathbb{N}$ a strictly increasing function, called the *reindexing* of our sequence. Equivalently we write $\tilde{\boldsymbol{x}} = \left\{x_{n_k}\right\}_k$ where $n_k = n(k)$.

We say a sequence is *bounded* if there exists an $M \in \mathbb{R}$ such that $d(x_n, x_m) \leq M$ for all $n, m \in \mathbb{N}$ (this is equivalent to saying there is some $M' \in \mathbb{R}$ and $x \in X$ such that $d(x_n, x) \leq M'$ for all $n \in \mathbb{N}$).

Definition 3 (Cauchy): A *Cauchy sequence* in a metric space (X, d) is a sequence $\{x_n\}$ such that

$$\forall\, \varepsilon > 0, \exists\, M \in N \text{ s.t. } n, m \geq M \Rightarrow d(x_n, x_m) < \varepsilon.$$

Proposition 1 (Some Properties of Sequences): Fix (X, d) a metric space and $\{x_n\}$ a sequence in X.

- $\{x_n\}$ convergent $\Rightarrow$ $\{x_n\}$ Cauchy. If the converse holds for every sequence in X, we say X is *complete*
- if $\{x_n\}$ Cauchy and has a converging subsequence, then $\{x_n\}$ converges along the whole sequence
- if $\{x_n\}$ Cauchy, $\{x_n\}$ bounded
- if $\{x_n\}$ converges, then every subsequence of $\{x_n\}$ converges, and to the same limit

Definition 4 (Continuity): A function $f : (X, d) \to (Y, \rho)$ is said to be continuous at $x \in X$ if for every $\varepsilon > 0$ there exists $\delta = \delta(x) > 0$ such that

$$d(x, x') < \delta \Rightarrow \rho(f(x), f(x')) < \varepsilon.$$

It is said to be *uniformly continuous* on X if δ as in the definition of continuity can be chosen independently of x.

Definition 5 (Absolute Continuity): Let $f : [a, b] \to \mathbb{R}$. We say f is *absolutely continuous* on $[a, b]$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any collection of disjoint intervals $(a_k, b_k) \subset [a, b]$, $k = 1, \ldots, N$ with $\sum_{k=1}^{N}(b_k - a_k) < \delta$, then $\sum_{k=1}^{N}|f(b_k) - f(a_k)| < \varepsilon$.

Proposition 2: $f : (X, d) \to (Y, \rho)$ is continuous at $x \in X$ iff for every $\{x_n\} \subset X$ such that $x_n \to x$ in X, $f(x_n) \to f(x)$ in Y.

Proposition 3: Let $f : (a, b) \to \mathbb{R}$ be a uniformly continuous function on the bounded open interval f. Then there exists a unique continuous extension $\tilde{f} : [a, b] \to \mathbb{R}$ that agrees with f on (a, b).

Proof. If it were to exist, by continuity, $\tilde{f}(a) = \lim_{x\downarrow a} f(a)$. Take any sequence x_n in (a, b) converging to a. One uses the uniform continuity to show that $\{f(x_n)\}$ is Cauchy and therefore converges since $\mathbb{R}$ complete. So this limit exists; moreover, one gets uniqueness in a similar vein by the well-definedness of this quantity (so basically continuity and uniqueness of limits). ■

Definition 6 (Normed and Inner Product Spaces): Let V be a vector space over $K \in \{\mathbb{R}, \mathbb{C}\}$. Then $\|\cdot\| : V \to \mathbb{R}$ is called a *norm* if:

- $\|v\| \geq 0 \,\forall\, v \in V$, and equal to zero iff $v = 0$
- $\|\alpha v\| = |\alpha| \; \|v\|$ for all $\alpha \in K, v \in V$
- $\|u + v\| \leq \|u\| + \|v\| \,\forall\, u, v \in V$

A function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ is called an *inner product* if, for all $u, v, w \in V$ and $\alpha \in K$:

- $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
- $\langle \alpha u, v \rangle = \overline{\alpha \langle u, v \rangle}$
- $\langle u, v \rangle = \overline{(\langle v, u \rangle)}$
- $\langle u, u \rangle \geq 0$ and equals zero iff $u = 0$,

Accordingly, $(V, \|\cdot\|)$ is called a *normed vector space* and $(V, \langle \cdot, \cdot \rangle)$ an *inner product space.*

Remark 1: Norms induce natural metrics, $d(u, v) := \|u - v\|$. Inner products induce natural norms, $\|u\| = (\langle u, u \rangle)^{1/2}$. So every inner product space is (naturally) a normed vector space, and every vector space is (naturally) a metric space. We say things like "V is a complete normed vector space" to mean V is complete as a metric space with the natural metric induced by its norm, for convenience.

Proposition 4 (Properties of Inner Product Spaces): Let V an inner product space.

- $|\langle u, v \rangle| \leq |u\|v|$

Definition 7 (Series): Let $(V, \|\cdot\|)$ be a normed vector space. A *series* in V is a sequence of partial sums, i.e. a sequence

$$S_n := \sum_{k=0}^{n} a_k$$

where $\{a_k\}$ a sequence in V. We say the series converges if $\{S_n\}$ converges in V.

Remark 2:

Theorem 1 (Banach Fixed Point): Let (X, d) a complete metric space. A map $T : X \to X$ is said to be a *contraction mapping* if there exists a constant $\alpha \in (0, 1)$ such that for all $x, y \in X$, $d(T(x), T(y)) \leq \alpha d(x, y)$.

If $T : X \to X$ a contraction, there exists a unique fixed point of T, i.e. a unique $x_0 \in X$ such that $T(x_0) = x_0$. Moreover, if T^n represents n-fold composition of T with itself, then $T^n(x) \to x_0$ as $n \to \infty$ for all $x \in X$.

Proof. (Uniqueness) If x_0, x_1 are two fixed points, then $d(T(x_0), T(x_1)) = d(x_0, x_1)$ on one hand, and by the contraction property, $d(T(x_0), T(x_1)) \leq \alpha d(x_0, x_1)$. Thus $d(x_0, x_1) \leq \alpha d(x_0, x_1)$, but $0 < \alpha < 1$ so this is only possible if $d(x_0, x_1) = 0$ i.e. if $x_0 = x_1$.

(Existence) Fix $x \in X$. Define the sequence $x_1 = x$ and $x_n = T^n(x) = T(x_{n-1})$ for $n \geq 2$. One shows this is a Cauchy sequence by using the contraction property (inductively; one picks up a sum of the form $1 + \alpha + \alpha^2 + \cdots$ which can be bounded by a geometric series). By completeness, x_n converges to some $x_0 \in X$. We claim x_0 is a fixed point. Indeed, one easily sees T is continuous, so $T^n(x) \to x_0$ implies $T(T^n(x)) \to T(x_0)$. On the other hand, $T(T^n(x)) = T^{n+1}(x) \to x_0$. By uniqueness, $T(x_0) = x_0$. ∎

1.1.2 Differentiation

Definition 8: Given $f : \mathbb{R}^n \to \mathbb{R}^m$, the derivative at $x_0 \in \mathbb{R}^n$ is the linear map $\mathrm{D}f(x_0) : \mathbb{R}^n \to \mathbb{R}^m$, if it exists, such that

$$\frac{\|f(x) - f(x_0) - Df(x_0)(x - x_0)\|}{\|x - x_0\|} \to 0 \text{ as } x \to x_0.$$

If we fix bases for $\mathbb{R}^n, \mathbb{R}^m$, then $Df(x_0)$ can be represented by a $m \times n$ matrix $J_f(x_0)$ called the *Jacobian* of f at x_0.

Remark 3: When $m = 1$, one often writes $\mathrm{D}f = (\nabla f)^\top$ where ∇f is viewed as a column vector in $\mathbb{R}^n$.

Proposition 5 (Properties of Derivative): Let $f, g : \mathbb{R}^n \to \mathbb{R}^m$ and $h : \mathbb{R}^k \to \mathbb{R}^n$, and $\alpha, \beta \in \mathbb{R}$.

- $\mathrm{D}f$ is unique when it exists
- $\mathrm{D}(\alpha f + \beta g) = \alpha\,\mathrm{D}f + \beta\,\mathrm{D}g$
- $\mathrm{D}(fg) = (\mathrm{D}f)g + f(\mathrm{D}g)$, when $m = 1$
- $\mathrm{D}(f \circ h) = (\mathrm{D}f)|_h \cdot (\mathrm{D}h)$ (*chain rule*)
- f differentiable at $x_0 \Rightarrow f$ (Lipschitz) continuous at x_0
- If $\mathrm{D}f$ exists at a point x_0, f partially differentiable in all directions, and

$$J_f(x_0) = \left(\frac{\partial f_j}{\partial x_i}(x_0)\right)_{\substack{1\leq i\leq n \\ 1\leq j\leq m}}$$

- If all of the partial derivatives of f exist and are continuous at x_0, then f differentiable at x_0

Remark 4: One need be careful with the last two properties - existence of partial derivatives needn't imply differentiability in general (we need some extra condition, such as continuity as stated here); consider

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x,y) = \begin{cases} \frac{x^2 y}{x^2+y^2} & (x,y) \neq 0 \\ 0 & (x,y) = 0 \end{cases}.$$

One can check $\partial_x f, \partial_y f$ both exist (and equal 0) at the origin, but f not differentiable at 0. Intuitively, $\mathrm{D}f$ captures "average change in all directions" while partial derivatives capture "change in a specified direction".

Definition 9 (Other Derivatives): Let $f : \mathbb{R}^n \to \mathbb{R}$ and $F = (F_1, \ldots, F_n) : \mathbb{R}^n \to \mathbb{R}^n$.

- $\frac{\partial}{\partial x_i} f(x_1, \ldots, x_n) := \lim_{h\to 0} \frac{f(x_1,\ldots,x_i+h,\ldots,x_n) - f(x_1,\ldots,x_n)}{h}$ is called the *partial derivative* of f in the x_i direction.
- Given a vector $d \in \mathbb{R}^n$ and $x \in \mathbb{R}^n$, define the *directional derivative* of f in the direction d at x by

$$\mathrm{d}f(x; d) := \lim_{h\to 0} \frac{f(x + hd) - f(x)}{h}.$$

- Define the *divergence* of F at x (assuming its first-partial derivatives exist) by

$$\mathrm{div}\ F(x) := \frac{\partial F_1}{\partial x_1}(x) + \cdots + \frac{\partial F_n}{\partial x_n}(x).$$

- If $n = 3$, define the *curl* of F at x (assuming its first-partial derivatives exist) by

$$\begin{aligned} \mathrm{curl}\ F(x) &:= \det\left(\begin{pmatrix} e_1 & e_2 & e_3 \\ \partial_x & \partial_y & \partial_z \\ F_1(x) & F_2(x) & F_3(x) \end{pmatrix}\right) \\ &= (\cdots) \end{aligned}$$

Proposition 6: If $f : \mathbb{R}^n \to \mathbb{R}$ differentiable at $x \in \mathbb{R}^n$, then

$$\mathrm{d}f(x; d) = \nabla f(x)^T.$$

If $d = e_i$ and f partially differentiable in x_i at x, then $\mathrm{d}f(x; d) = \frac{\partial f}{\partial x_i}(x)$.

Theorem 2 (Mean-Value Theorem): Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable on an open ball $B \subset \mathbb{R}^n$. Then for any $x, y \in B$, there exists a $z \in B$ such that

$$f(x) - f(y) = \mathrm{D}f(z_0)(x - y).$$

Proof. Define $\varphi(t) := f(y + t(x-y))$ for $t \in [0,1]$, noting that $\varphi(0) = y$ and $\varphi(1) = x$. One computes, using the chain rule, that $\varphi'(t) = \mathrm{D}f(y + t(x - y)) \cdot (x - y)$. By the one-variable mean-value theorem, there exists some $t_0 \in [0,1]$ such that $\varphi(1) - \varphi(0) = \varphi'(t_0)$. Letting $z = y + t_0(x - y)$ gives the desired result, noting that z indeed in B by convexity.

For completeness, we also prove the one-variable mean-value theorem. It suffices to prove that if $\varphi : \mathbb{R} \to \mathbb{R}$ is differentiable and is such that $\varphi(a) = \varphi(b) = 0$ then there exists $c \in (a,b)$ such that $\varphi'(c) = 0$, since for a general function f we can define $\varphi(x) := f(x) - \varphi(a)$ ■

Theorem 3 (Mean-Value Theorem, Integral Form): Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. Then for any $x, y \in \mathbb{R}^d$,

$$f(y) - f(x) = \int_0^1 \nabla f(x + t(x - y)) \cdot (x - y)\,\mathrm{d}t.$$

Proof. Let $\psi(t) = f(x + t(x - y))$. Note that $\psi'(t) = \nabla f(x + t(x - y)) \cdot (x - y)$, so by fundamental theorem of calculus,

$$\int \nabla f(x + t(x - y)) \cdot (x - y)\,\mathrm{d}t = \int_0^1 \psi'(t)\,\mathrm{d}t = \psi(1) - \psi(0) = f(y) - f(x).$$

■

Theorem 4 (Clairaut's Theorem): Suppose $f : \mathbb{R}^n \to \mathbb{R}$ twice differentiable at x_0. Then, $\partial_{x_j} f(x_0) = \partial_{x_i} f(x_0)$ for all $i, j = 1, \ldots, n$.

Proof. *(Sketch)* For $n = 2$ (the proof extends to general dimensions by fixing all coordinates except the ith, jth) in variables (x, y), consider the function

$$\varphi(s, t) := f(x_0 + s, y_0 + t) - f(x_0 + s, y_0) - f(x_0, y_0 + t) + f(x_0, y_0).$$

The idea is to recognize $\varphi(s, t)$ as a difference of functions in the s, t variables resp., apply mean-value theorem to pick-up $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ terms, then use the definition of the second-order partial derivatives to estimate φ in terms of $\frac{\partial^2}{\partial x \partial y}, \frac{\partial^2}{\partial y \partial x}$. One should ultimately find that $\frac{\varphi(t,t)}{t^2} \to \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0)$ and $\frac{\partial^2 f}{\partial y \partial x}$, depending on which variable one chooses to expand in first. ■

Theorem 5 (Inverse Function Theorem): Let $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ a C^k map such that $\mathrm{D}F(x_0)$ invertible for some $x_0 \in \Omega$. Then there exist neighborhoods $U \subset \Omega$ of x_0 and $V \subset \mathbb{R}^n$ of $F(x_0)$ and a C^k map $G : V \to U$ such that

$$(F \circ G)(y) = y \quad \forall\, y \in V, \qquad (G \circ F)(x) = x \quad \forall\, x \in U.$$

Moreover, $\mathrm{D}F(x)$ invertible for $x \in U$, and

$$\mathrm{D}G(y) = (\mathrm{D}F)^{-1}(x), \quad y = F(x), x \in U.$$

Theorem 6 (Implicit Function Theorem): Let $F : \Omega := \Omega_x \times \Omega_y \subset \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be a C^k map, and assume $X_0 = (x_0, y_0) \in \Omega$ is such that $F(X_0) = 0$ and $\mathrm{D}_y F(X_0)$ is invertible (*where* D_y *the derivative in just the* y*-variables, i.e. represented by a* $m \times m$ *matrix*). Then, there exists a neighborhood $U \subset \Omega_x$ of x_0 and a C^k function $h : \Omega_x \to \mathbb{R}^m$ such that

$$F(x, h(x)) = 0, \quad \forall\, x \in \Omega.$$

Proof. *(Sketch)* Define $f : \Omega \to \mathbb{R}^n \times \mathbb{R}^m$ by $f(X) := (X, F(X))$ ("augmenting" F to map to and from spaces of the same dimension). One checks

$$J_f = \begin{pmatrix} I_{n\times n} & 0_{n\times m} \\ \mathrm{D}_x F & \mathrm{D}_y F \end{pmatrix},$$

which has determinant $\det(\mathrm{D}_y F)$ using properties of block matrices. This is non-zero at X_0 by assumption, so we may apply the inverse-function theorem, so locally there exists an inverse g. If we write $g = (\tilde{g}, \tilde{h}) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^{n+m}$, we see that $f(g(x, y)) = (x, y)$ (by inverse property) and $f(g(x, y)) = \tilde{g}(x, y), F(\tilde{g}(x, y), \tilde{h}(x, y))$ by definition. In particular we see that $\tilde{g}(x, y) \equiv x$, and thus we get the property

$$F(x, \tilde{h}(x, y)) = y.$$

Thus the map $h(x) := \tilde{h}(x, 0)$ gives the desired function. ■

Theorem 7 (Lagrange Multipliers, one constraint): Let $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be C^1 and $\Sigma := \{x \in \Omega : g(x) = 0\}$, and assume $\mathrm{D}g(x)$ nonzero (i.e. the gradient of g) on all of Ω. Then, if f restricted to Σ has a max/min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda g(x_0).$$

Proof. Let x_0 be a max/min of f restricted to Σ. Assume wlog that $\frac{\partial}{\partial x_n} g(x_0) \neq 0$. By implicit function theorem, then locally to x_0, Σ is represented by $\{(x', h(x'))\}$ (where $x' \in \mathbb{R}^{n-1}$) for some C^1 function $h : \mathbb{R}^{n-1} \to \mathbb{R}$. Defining $F : \mathbb{R}^{n-1} \to \mathbb{R}$, $F(x') := f(x', h(x'))$, this implies $(x_0', h(x_0'))$ a local max/min of F and thus it's gradient must equal zero there. Chain rule gives

$$\nabla F(x') = \left(\frac{\partial}{\partial x'_i} f(x', h(x')) + \frac{\partial}{\partial x_n} f(x', h(x')) \frac{\partial h(x')}{\partial x_i}\right)_{i=1}^{n-1}. \quad (\dagger)$$

Also since $g(x', h(x')) = 0$ locally to x_0, we can take the derivative of this expression and find

$$0 = \left(\frac{\partial g}{\partial x_i}(x_0) + \frac{\partial h}{\partial x'_i}(x_0) \frac{\partial g}{\partial x_n}(x_0)\right)_{i=1}^{n-1}. \quad (\ddagger).$$

Taking $x' = x'_0$ in $(\dagger)$ makes the right-hand side 0. Equating $(\dagger), (\ddagger)$ and simplifying like-terms yields the proof, where λ is found to be $\left(\left[\frac{\partial f}{\partial x_n}\right] / \left[\frac{\partial g}{\partial x_n}\right]\right|_{x_0}$. ∎

Remark 5: When we say "Σ locally looks like V", what we really mean is there exists an open set U such that $\Sigma \cap U = V$.

Remark 6: In particular, then if f, g are C^1 on some open superset of Ω, and $\partial\Omega = \{x \in \Omega : g(x) = 0\}$, then one can compute global maxima of f by
(i) computing *stationary points* i.e. such that $\nabla f = 0$, inside Ω, and
(ii) computing points such that $\nabla f = \lambda \nabla g$, on $\partial\Omega$.

Then, if x_0 a global max of f, then either it lies inside Ω in which case $\nabla f(x_0) = 0$, or it lies on the boundary in which case there exists λ such that $\nabla f(x_0) = \lambda \nabla g(x_0)$. Thus, the global max of f is the point of greatest function value among those from 1., 2.. A similar statement holds for the global min.

Theorem 8 (Lagrange Multiplier, multiple constraints): Let $f, g_1, ..., g_k : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be C^1. Let $\Sigma = \{x \in \Omega : g_i(x) = 0 \,\forall\, i = 1, ..., k\}$, and assume $\{\nabla g_i(x)\}_{i=1}^k$ a linearly independent set of vectors for all $x \in \Sigma$. Then if f restricted to Σ has a local max/min at $x_0 \in \Sigma$, then there exists $\lambda_1, ..., \lambda_k$ such that

$$\nabla f(x_0) = \lambda_1 \nabla g_1(x_0) + \cdots + \lambda_k \nabla g_k(x_0).$$

Theorem 9 (Taylor's Theorem): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be a $C^{k+1}(\Omega)$ function. For any $x, x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^{k} \sum_{1 \le i_1, ..., i_j \le n} \frac{\partial^j f}{\partial x_{i_1} \cdots \partial_{x_{i_j}}} \left(x_{i_1} - x^0_{i_1}\right) \cdots \left(x_{i_j} - x^0_{i_j}\right) + R_k(x, x^0),$$

where the *remainder* R_k is such that

$$\left|R_k(x, x^0)\right| \le M_k \left|x - x^0\right|^{k+1} \quad \text{as } x \to x^0,$$

where M_k depends only on the $(k+1)$-st partial derivatives of f.

Remark 7: The easiest way to prove this is to apply 1-dimensional Taylor's theorem to the function $\varphi(t) := f\left(x^0 + t\frac{x - x^0}{\|x - x^0\|}\right), \quad t \in \left[0, \|x - x^0\|\right]$, for $t$sufficiently small, and expanding the derivatives of φ by chain rule.

On the other hand, the easiest way to derive the 1-diemnsional Taylor's theorem is by repeated application of the fundamental theorem of calculus,

$$\begin{aligned}
\varphi(t) &= \varphi(t_0) + \int_{t_0}^{t} \underbrace{\varphi'(s)}\, \mathrm{d}s \\
&= \varphi(t_0) + \int_{t_0}^{t} \left[\varphi'(t_0) + \int_{t_0}^{s} \varphi''(u)\, \mathrm{d}u\right] \mathrm{d}s \\
&= \varphi(t_0) + \varphi'(t_0)(t - t_0) + \int_{t_0}^{t} \int_{t_0}^{s} \varphi''(u)\, \mathrm{d}u\, \mathrm{d}s,
\end{aligned}$$

etc. In particular, one sees that this implies the remainder term can be written as

$$R_k(t) = \int_{t_0}^{t} \int_{t_0}^{s_1} \cdots \int_{t_0}^{s_{k-1}} \varphi^{(k)}(u)\, \mathrm{d}u \cdots \mathrm{d}s_2\, \mathrm{d}s_1,$$

from which obtaining the bound on R_k follows by approximating this integral.

Theorem 10 (Taylor's Theorem, Weaker Assumptions): Let $f \in C^{k-1}(\Omega)$ and such that $\mathrm{D}^{k-1} f$ is differentiable at $x^0 \in \Omega$. Then for $x \in \Omega$, the same conclusion as above holds, but with remainder

$$\lim_{x \to x^0} \frac{R_k(x)}{\left|x - x^0\right|^k} = 0,$$

i.e. we can't in general say anything about how *quickly* this quantity goes to zero.

Remark 8: In "asymptotic" notation, the first theorem says $R_k(x) = \mathcal{O}\left(\left|x - x^0\right|^{k+1}\right)$ and the second says $R_k(x) = o\left(\left|x - x^0\right|^k\right)$.

1.1.3 Integration

Definition 10 (Partition): Given a rectangle $R \subset \mathbb{R}^n$, a *box partition* of R is a **finite** collection $\mathcal{P} = \{B_k\}_{k=1}^N$ of *boxes* (rectangles) B_k such that $R = \bigcup_{k=1}^N B_k$.

Definition 11 (Upper/Lower Riemann Sum): Let $f : R \subset \mathbb{R}^n \to \mathbb{R}$ a bounded function. Given a partition $\mathcal{P}$ of R, define the *upper, lower Riemann sums*

$$U(f, \mathcal{P}) := \sum_{B \in \mathcal{P}} \sup_{x \in B} f(x) \cdot \mathrm{vol}(B), \quad L(f, \mathcal{P}) := \sum_{B \in \mathcal{P}} \inf_{x \in B} f(x) \cdot \mathrm{vol}(B),$$

where, if $B = [a_1, b_1] \times \cdots \times [a_n, b_n]$, $\mathrm{vol}(B) := (b_1 - a_1) \cdots (b_n - a_n)$. Define the *upper, lower Riemann integrals* of f by

$$\overline{\int_R} f \,\mathrm{d}x := \inf_{\mathcal{P}:\text{ partition of } R} U(f, \mathcal{P}), \qquad \underline{\int_R} f \,\mathrm{d}x := \sup_{\mathcal{P}:\text{ partition of } R} L(f, \mathcal{P}).$$

In particular, we trivially have the inequalities

$$\inf_R f \cdot \mathrm{vol}(R) \leq L(f, \mathcal{P}) \leq \underline{\int_R} f \,\mathrm{d}x \leq \overline{\int_R} f \,\mathrm{d}x \leq U(f, \mathcal{P}) \leq \sup_R f \cdot \mathrm{vol}(R).$$

If the lower, upper Riemann sums agree, we write their shared value by $\int_R f \,\mathrm{d}x$, called the Riemann integral of f over R.

Remark 9: We can extend this definition to general f over bounded domains Ω by taking R a rectangle such that $R \supset \Omega$ and extending f by zero outside of R.

Definition 12 (Content Zero): A set $\Omega \subset \mathbb{R}^n$ is said to be of *content zero* if for every $\varepsilon > 0$, there exists a finite number of boxes $B_1, \ldots, B_N$ which cover Ω and such that $\sum_{n=1}^N \mathrm{vol}(B_n) \leq \varepsilon$.

Theorem 11: If f bounded on R and has set of discontinuities with content 0, it is Riemann integrable on R.

Proposition 7: Riemann integrals are linear.

Theorem 12 (Fubini):

Definition 13 (Regular Partition): Let $\Omega \subset \mathbb{R}^n$ a domain with $\mathrm{content}(\partial\Omega) = 0$. A *regular partition* $\mathcal{P} = \{D_k\}_{k=1}^N$ is a finite collection of sets D_k satisfying

- $\Omega = \bigcup_{k=1}^N D_k$ and $\mathrm{content}(\partial D_k) = 0$ for all $k = 1, \ldots, N$, and
- $\mathrm{content}\left(D_k \cap D_j\right) = 0$ for all $k \neq j$.

Proposition 8: Let $f : \Omega \to \mathbb{R}$. Then f is Riemann integrable over Ω iff for all $\varepsilon > 0$ there exists a *regular* partition $\mathcal{P}$ such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$

Theorem 13 (Change of Variables): Let $\Omega, \Omega' \subset \mathbb{R}^n$ be bounded domains and $T : \Omega' \to \Omega$ a C^1 bijection. Then, for any $f : \Omega \to \mathbb{R}$ which is Riemann integrable over Ω, then

$$\int_\Omega f \,\mathrm{d}V = \int_{\Omega'} (f \circ T) \, |\det \mathrm{D}T| \,\mathrm{d}V.$$

PROOF. ■

Theorem 14 (Differentiation under the Integral):

Theorem 15 (Improper Integrals):

1.1.4 Vector Calculus

Definition 14 (Curve): A *(parametrized) C^k curve* in $\mathbb{R}^n$ is a connected set $\mathcal{C} \subset \mathbb{R}^n$ such that there exists a C^k function $\gamma : I \to \mathbb{R}^n$ with $\gamma(I) = \mathcal{C}$, where $I \subset \mathbb{R}$ an interval. We say $\mathcal{C}$ *regular* if γ' never vanishes on I.

A *piecewise C^k curve* is a set $\mathcal{C} = \mathcal{C}_1 \cup \cdots \cup \mathcal{C}_m$ where each $\mathcal{C}_i$ a C^k curve and each has "shared endpoints" (in practice, think of something like a triangle).

Remark 10 : More generally, a curve is generally defined as a connected subset of $\mathbb{R}^n$ that is “locally a parametrized curve”.

Definition 15 (Surface): A C^k *surface* in $\mathbb{R}^n$ is a connected set $S \subset \mathbb{R}^n$ such that for every point $p \in S$, there exists an open subset $U \subset \mathbb{R}^2$ and $V \subset \mathbb{R}^n$ such that $p \in V$, and a C^k homeomorphism $\varphi : U \to V \cap S$ such that $\varphi(U) = V \cap S$.

We call S regular if every such φ has $\mathrm{d}\varphi_q : \mathbb{R}^2 \to \mathbb{R}^n$ is one-to-one for all $q \in U$, i.e. the Jacobian $J_\varphi(q)$ has rank 2.

Definition 16 (Tangent Plane): Let $S \subset \mathbb{R}^3$ a regular C^1 surface. The *tangent plane* at a point $p \in S$ is

$$T_pS = \mathrm{d}_q\varphi(\mathbb{R}^2),$$

where $\varphi(q) = p$. Since $\mathrm{d}_q\varphi$ one-to-one, this is a 2-dimensional subspace of $\mathbb{R}^3$.

Remark 11 : This is well-defined, regardless of choice of parametrization, by an inverse function theorem argument.

Definition 17 (Normal): A normal to a regular C^1 surface $S \subset \mathbb{R}^3$ at a point p is a vector $n \in \mathbb{R}^3$ such that n is orthogonal to T_pS. If we restrict n to be of unit length, then since T_pS two dimensional, there are precisely two such n's, given by $\pm(v_1 \times v_2)$ where $\{v_1, v_2\}$ a basis for T_pS.

Definition 18 (Integration over a Surface in $\mathbb{R}^3$): Let $S \subset \mathbb{R}^3$ a regular parametrized surface with parametrization $\varphi : D \to \mathbb{R}^3$ and $f : \mathbb{R}^3 \to \mathbb{R}$ Riemann integrable. Define the *surface integral*

$$\iint_S f \,\mathrm{d}\sigma := \iint_D (f \circ \varphi)\, |\varphi_u \times \varphi_v| \,\mathrm{d}u\,\mathrm{d}v.$$

Suppose $F : \mathbb{R}^3 \to \mathbb{R}^3$ a vector field, and S an oriented surface with unit normal n. Define the *flux integral*

$$\iint_S F \cdot \mathrm{d}\sigma := \iint_S (F \cdot n)\,\mathrm{d}\sigma = \iint_D (F \circ \varphi) \cdot (n \circ \varphi)\, |\varphi_u \times \varphi_v| \,\mathrm{d}u\,\mathrm{d}v.$$

Remark 12 : Since we are in dimension 3, $n = \pm\frac{\varphi_u \times \varphi_v}{|\varphi_u \times \varphi_v|}$ so the flux integral can be evaluated as $\pm \iint_D (F \circ \varphi) \cdot (\varphi_u \times \varphi_v)\,\mathrm{d}u\,\mathrm{d}v$ (one needs to be careful of using the right sign).

Theorem 16 (Green's): Let $F = (P, Q) : \mathbb{R}^2 \to \mathbb{R}^2$ a C^1 vector field. Let $D \subset \mathbb{R}^2$ a bounded region with piecewise C^1, positively oriented boundary ∂D. Then,

$$\iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}A = \int_{\partial D} F \cdot \mathrm{d}s.$$

Theorem 17 (Stoke's): Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ a C^1 vector field and S a regular surface in $\mathbb{R}^3$ with piecewise C^1 boundary ∂S. Then,

$$\iint_S \mathrm{curl}(F) \cdot \mathrm{d}S = \int_{\partial S} F \cdot \mathrm{d}s.$$

Theorem 18 (Divergence): Let $F : \mathbb{R}^n \to \mathbb{R}$ a C^1 function and $W \subset \mathbb{R}^n$ a solid bounded region with boundary $S = \partial W$ which is a regular C^1 surface. Then,

$$\int_W \mathrm{div}\ F \,\mathrm{d}V = \int_S F \cdot \mathrm{d}S.$$

1.1.5 Analysis on Functions

Remark 13 : A lot of what is stated in this section could be generalized without too much hassle to functions on/to general metric spaces, but is cumbersome and unnecessary for my interests.

Definition 19 (Convergence of sequences of functions): Let $\{f_n\}$ be a sequence of real-valued functions defined on some set S. We say $f_n \to f$ *pointwise on S* if $f_n(x) \to f(x)$ (as sequences of real-numbers) for every $x \in S$.

We say that the convergence is *uniform* if $\sup_{x \in S} |f_n(x) - f(x)| \to 0$ as $n \to \infty$, or equivalently if $f_n \to f$ under the *uniform norm* $\|f\|_\infty := \sup_S |f|$ (where we may omit the set if clear from context).

Remark 14: Equivalently, $f_n \to f$ uniformly if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in S$.

Remark 15: It's clear that uniform $\Rightarrow$ pointwise convergence. A good counterexample for the other direction is the functions $f_n(x) = x^n$ defined on $[0, 1]$. We see that f_n converges pointwise to the function that is 0 everywhere except 1, at which point it is equal to 1. However, this convergence is not uniform.

Theorem 19: Let $C(K)$ be the set of continuous functions on a compact set $K \subset \mathbb{R}^n$. This is a complete metric space when equipped with the uniform norm.

Remark 16: In particular, this means that the uniform limit of continuous functions is continuous.

Theorem 20 (Interchange of Limit and Integral): Let $\{f_n\}$ be a sequence of Riemann integrable functions on some box domain B which converge uniformly to some Riemann integrable function f on B. Then, $\int_B f_n \,\mathrm{d}x \to \int_B f \,\mathrm{d}x$.

Proof. $|\int_B f \,\mathrm{d}x - \int_B f_n \,\mathrm{d}x| \leq \int_B |f - f_n| \,\mathrm{d}x \leq \mathrm{vol}(B) \sup_B |f - f_n|$; the right-hand side above goes to zero by uniform convergence. ■

Theorem 21 (Interchange of Limit and Derivative): Suppose $f : U \subset \mathbb{R}^n \to \mathbb{R}$ where U open, where f_m differentiable, converge pointwise at some point $x_0 \in U$, and their derivatives converge uniformly, then $\{f_m\}$ converges uniformly to some differentiable function f and $\partial_x f_m(x) \to \partial_x f(x)$ as $m \to \infty$ on U.

Proof. Assume $n = 1$ for simplicity. Let $g_n = f_n'$ and let g be the uniform limit of g_n. Take as candidate $f(x) := f(x_0) + \int_{x_0}^x g(y) \,\mathrm{d}y$. It's clear f is differentiable, and by the fundamental theorem of calculus,

$$\begin{aligned}
|f_n(x) - f(x)| &= |f_n(x_0) - f(x_0) + \int_{x_0}^x f_n'(y) - g(y) \,\mathrm{d}y\ | \\
&\leq |f_n(x_0) - f(x_0)| + \int_{x_0}^x |f_n'(y) - g(y)| \,\mathrm{d}y \\
&\leq |f_n(x_0) - f(x_0)| + |x - x_0| \sup_y |f_n'(y) - g(y)|,
\end{aligned}$$

and the right-hand side converges by the point-wise convergence at x_0 and the uniform convergence of the derivatives. Finally, for $x \in U$, $|f_n'(x) - f'(x)| = |g(x) - g_n(x)|$ by definition, so the convergence is immediate. ■

Theorem 22 (Bounded Convergence for Integrals): Assume $f_n \to f$ pointwise, $\{f_n\}, f$ Riemann integrable, and $|f_n(x)| \leq B$ for all $x \in I, n \geq 1$. Then $\int_I f_n(x) \,\mathrm{d}x \to \int_I f(x) \,\mathrm{d}x$.

Proof.

■

Theorem 23 (Dini's Theorem): Let $\{f_n\}, f$ continuous functions on I and $f_n \to f$ pointwise. Suppose $f_n(x) \leq f_{n+1}(x)$ for all x and $n \geq 1$. Then $f_n \to f$ uniformly.

Theorem 24 (Interchange of Integral and Summation): Suppose $f_n : K \to \mathbb{R}$ are Riemann integrable and $\sum_n f_n \to f$ uniformly, and f Riemann integrable. Then,

$$\sum_n \int_K f_n \,\mathrm{d}x = \int_K \sum_n f_n \,\mathrm{d}x.$$

Proof. Letting $S_N(x) = \sum_{n \leq N} f_n(x)$, this theorem says $\lim_N \int_K S_N \,\mathrm{d}x = \int_K \lim_N S_N \,\mathrm{d}x$. This follows from an earlier proposition on uniform convergence of sequences of functions and integrals. ■

Theorem 25 (Power Series): A *power series* (centered at x_0) is a function of the form

$$f(x) = \sum_{n \geq 0} a_n (x - x_0)^n.$$

Define $\frac{1}{R} := \limsup_{n \to \infty} |a_n|^{1/n}$ (taking $\frac{1}{R} = 0, \infty \leftrightarrow R = \infty, 0$ resp. as convention). Then,

- if $|x - x_0| < R$, the series converges absolutely (and in fact uniformly over compact sets)
- if $|x - x_0| > R$, the series diverges

Proof. Compare to a geometric series. ■

Theorem 26 (Differentiating, Integrating term-by-term): Let $f(x) = \sum_{n \geq 0} a_n (x - x_0)^n$ have positive radius of convergence. Then, f is infinitely differentiable and integrable on its interval of convergence, obtained by differentiating/integrating term-by-term, i.e.

$$f'(x) = \sum_{n \geq 0} n a_n (x - x_0)^{n-1},$$

$$\int f(x) \, \mathrm{d}x = C + \sum_{n \geq 0} \frac{a_n}{n+1} (x - x_0)^{n+1}.$$

Theorem 27 (Weierstrass Approximation Theorem): Let $I \subset \mathbb{R}$ a closed and bounded interval and $f \in C^0(I)$ a continuous function. Then, for every $\varepsilon > 0$ there exists a polynomial p_ε such that

$$\|f - p_\varepsilon\|_\infty < \varepsilon.$$

That is, continuous functions are dense in the space of continuous functions on a compact interval with respect to the uniform norm.

Definition 20: Let (X, d) a metric space. A family of functions $\mathcal{F} \subset C(X)$ is said to be:

- *uniformly bounded* if $\sup_{(f \in \mathcal{F} \|f\|_\infty < \infty}$
- *equicontinuous* if for all $\varepsilon > 0$ and $x \in X$ there exists $\delta > 0$ such that $d(x, y) < \delta$ implies $\sup_{f \in \mathcal{F}} |f(x) - f(y)| < \varepsilon$; it is said to be *uniformly equicontinuous* if δ is independent of x

Theorem 28 (Arzela-Ascoli): Let (X, d) a compact metric space and $\mathcal{F} \subset C(X)$ be a uniformly bounded and uniformly equicontinuous family of functions. Then, $\mathcal{F}$ is precompact with respect to the uniform metric, that is, for any sequence $\{f_n\} \subset \mathcal{F}$, there exists a uniformly converging subsequence $\{f_{n_k}\}$.

Remark 17: Usually uniform boundedness is easy to check. Uniform equicontinuity actually follows from equicontinuity when X compact, so it suffices to check this. A sufficient condition for uniform equicontinuity is uniform Lipschitz continuity, i.e. there exists $L > 0$ such that for all $x, y \in X$,

$$\sup_{f \in \mathcal{F}} |f(x) - f(y)| \leq L d(x, y),$$

which is usually easier to establish if it holds.

1.2 Complex

1.2.1 Analytic Functions

Throughout this section, we'll usually assume $\Omega \subset \mathbb{C}$ a connected open set.

Definition 21 (Analytic/Holomorphic): A function $f : \mathbb{C} \to \mathbb{C}$ is said to be *analytic* on Ω if f is given by a converging power series everywhere in Ω. It is said to be holomorphic if f' exists.

For $z = x + iy \in \mathbb{C}$ and $f : \mathbb{C} \to \mathbb{C}$, write $f(z) = u(x, y) + v(x, y)i$ for real-valued functions u, v. Then, the *Cauchy-Riemann (CR) equations* are the equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Theorem 29 (Cauchy-Riemann Equations): f is holomorphic if and only if u, v satisfy the CR equations.

Proof. ($\Rightarrow$) follows by taking different limit directions in the definition of the derivative. ($\Leftarrow$) follows by writing, for some small $h = h_1 + h_2 i$,

$$u(x + h_1, y + h_2) = u(x, y) + h_1 \partial_x u(x, y) + h_2 \partial_y u(x, y) + |h| \; \psi_1(h), \quad \psi_1(h) \underset{h \to 0}{\to} 0,$$

similar for v. This implies, simplifying things with CR,

$$f(z + h) = f(x, y) + (\partial_x v - i\partial_y u)(h) + \psi(h)h|, \quad \psi(y) = o(|h|),$$

which gives the proof upon dividing both sides by h and sending $h \to 0$. ∎

Definition 22 (Contour Integration): Given a piecewise C^1 contour $C = C_1 \cup \cdots \cup C_k$ with C_i parametrized by C^1 functions $\gamma_i : [t_i, t_{i+1}] \to C_i$, define

$$\int_{C^{\pm}} f(z) \, \mathrm{d}z := \pm \sum_{i=1}^{k} \int_{t_i}^{t_{i+1}} (f \circ \gamma_i)(s) \cdot \gamma_i'(s) \, \mathrm{d}s,$$

where the $\pm$ indicates the *orientation* of C.

A domain Ω is said to be *simply connected* if for any two curves $\gamma_0, \gamma_1 : [0, 1] \to \mathbb{C}$ with common endpoints α, β in Ω, there exists a function $\gamma_t : I \to \mathbb{C}$ for $t \in [0, 1]$ such that $(t, s) \mapsto \gamma_t(s)$ is a continuous map, $\gamma_0 = \gamma_0$ and $\gamma_1 = \gamma_1$, and $\gamma_t(0) = \alpha, \gamma_t(1) = \beta$ for all $t \in [0, 1]$.

A curve C is said to be *simple* if it does not intersect itself, i.e. it has an injective parametrization, and *closed* if its parametrization $\gamma : I \to \mathbb{C}$ is equal on the endpoints of I.

Theorem 30 (Cauchy): Let f be a holomorphic function defined on Ω being simply connected with C a simple closed curve contained in Ω. Then

$$\int_C f(x) \, \mathrm{d}z = 0.$$

Partial converse:

Theorem 31 (Morera): Let f be continuous on Ω and such that

$$\int_\gamma f \, \mathrm{d}z = 0$$

for every closed, piecewise C^1 curve γ contained in Ω. Then f is holomorphic in Ω.

Theorem 32 (Cauchy's Integral Formula): Let f holomorphic on Ω, which contains a circle C and its interior. Then

$$f(z) = \int_{C^+} \frac{f(w)}{w-z}\,\mathrm{d}w, \quad \forall\, z \in \Omega.$$

In particular, this implies f is analytic on all of Ω, with

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_{C^+} \frac{f(w)}{(w-z)^{n+1}}\,\mathrm{d}w,$$

and thus

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n,$$

for any $z_0 \in \Omega$ and z such that $z - z_0 \in \Omega$.

Proof. Make a small indent in the contour around the singularity of the denominator and use holomorphicity to approximate. ■

Corollary 1 (Cauchy's Inequalities): Let $f : \Omega \to \mathbb{C}$ holomorphic. Then

$$|f^{(n)}(z)| \leq \frac{n!\|f\|_{C_R(z),\infty}}{R^n},$$

where $R > 0$ any real number such that $D_R(z) \subset \mathbb{C}$, where $D_R(z)$ the R-radius disc centered at z, and

$$\|f\|_{C_R(z),\infty} := \sup_{w \in C_r(z)} |f(w)|.$$

Corollary 2 (Liouville's Theorem): Let $f : \mathbb{C} \to \mathbb{C}$ be *entire*, i.e. holomorphic everywhere, and bounded. Then f is constant.

Proof. For every $R > 0$ and z fixed, Cauchy's inequalities given that $|f'(z)| \leq \frac{C}{R}$ for some C uniform in z. Letting $R \to \infty$ gives that $f'(z) = 0$ everywhere, and thus $f \equiv$ constant. ■

Theorem 33 (Analytic Continuation): Let f holomorphic on Ω, and assume either

- $f \equiv 0$ on a open set $D \subset \Omega$ or
- $f = 0$ on a sequence $\{z_n\} \subset \Omega$ and its limit point z_∞.

Then $f \equiv 0$ on Ω.

Remark 18: As a result, if two holomorphic functions agree on an open subset of Ω or on a converging sequence, then they agree everywhere in Ω. This is the *principle of analytic continuation*, demonstrating how rigid analytic functions are.

Proof. The first criteria follows from the first by a connectedness argument. The second follows by contradiction and looking at a local power series centered at z_∞. ■

Theorem 34 (Convergence of Holomorphic Functions): Let $\{f_n\}$ a sequence of holomorphic functions converging uniformly on compact subsets of Ω to some function f. Then f is holomorphic, and moreover, f_n' converges uniformly to f' on compact subsets.

Proof. This follows directly from Morera's theorem; the uniform convergence implies that $0 = \int_C f_n \,\mathrm{d}z \to \int_C f \,\mathrm{d}z = 0$. ■

Remark 19: The analgous statement is *not* true in real variables! That is, uniform convergence of differentiable real-valued functions are not necessarily differentiable.

1.2.2 Meromorphic Functions

Definition 23 (Singularities): A holomorphic function $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ is said to have a *singularity* at z_0. z_0 is called a

- *removable singularity* of f if there exists a holomorphic function $\tilde{f} : \Omega \to \mathbb{C}$ that agrees with f on $\Omega \setminus \{z_0\}$;
- *pole singularity* of f if $\frac{1}{f}$ has a removable singularity at z_0 (we define $\frac{1}{f}$ to be zero at z_0); or
- *essential singularity* of f otherwise.

We will say a singularity is *isolated* if there exists a neighborhood around it in which it is the only singularity of f.

Proposition 9 (Order of zeros, poles): Let $f : \Omega \to \mathbb{C}$ holomorphic have an isolated zero at $z_0 \in \Omega$. Then there exists a neighborhood of z_0, a unique positive integer $n := \operatorname{ord}_f(z_0)$, and a nonvanishing holomorphic function g such that $f(z) = (z - z_0)^n g(z)$ on that neighborhood.

The same statement holds if z_0 an isolated pole of f, with n a negative integer. When $n = -1$ we call the pole *simple;* more generally, $-n$ is called the *order/mulitiplicity* of z_0.

Proof. Follows from writing f in a local power series centered at z_0; there exists a minimally indexed nonzero coefficient. Factoring out this term gives the desired representation. Uniqueness follows easily. ■

Corollary 3: If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z) =: P(z) + G(z),$$

for some holomorphic (locally to z_0) function G.

The function P is called the *principal part* of f at z_0, and a_{-1} is called the *residue*, denoted $\operatorname{res}_{z_0} f = a_{-1}$.

Proof. Follows from the previous proposition by expanding terms. ■

Proposition 10 (Computation of Residues): If $n = \operatorname{ord}_f(z_0)$,

$$\operatorname{res}_{z_0} f = \lim_{z \to z_0} \frac{1}{(n-1)!} \left(\frac{\mathrm{d}}{\mathrm{d}z}\right)^{n-1} (z - z_0)^n f(z).$$

Proof. This follows from the previous corollary by taking computing the appropriate derivatives term-by-term. ■

Theorem 35 (Residue Theorem): Let f be holomorphic in an open set containing a disc with boundary C, except for finitely many poles $z_1, ..., z_N$ in the disc. Then,

$$\int_C f(z)\,\mathrm{d}z = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Proof. The idea is to use a "keyhole contour" that indents the original circle in N places to miss the poles. By holomorphicity, the integral over this contour is zero and one finds that (since the "walls" of the keyhole neighborhoods tend to zero) our integral in question is equal to the sum over the poles of the integral of f over small circles centered at each pole. Appealing to the representation in the corollary from earlier and computing quickly yields the result, using crucially the fact that

$$\int_C \frac{1}{(z - z_0)^n}\,\mathrm{d}z = \begin{cases} 0 & n > 1 \\ 2\pi i & n = 1 \end{cases}.$$

■

Theorem 36 (Classification of Isolated Singularities): Suppose $f : \Omega \setminus \{z_0\} \to \mathbb{C}$ holomorphic. Then:

- if f bounded on $\Omega \setminus \{z_0\}$, z_0 a removable singularity
- z_0 a pole iff $|f(z)| \to \infty$ as $z \to z_0$.

Proof. The second follows from the first. For the first, one contends that, inspired by the Cauchy's integral formula, the representation $\frac{1}{2\pi i}\int_C \frac{f(w)}{w-z}dw$ remains a holomorphic function that agrees with f everywhere away from z_0. ■

Proposition 11 (Residue at Infinity and Sum of Residues): Let f be holomorphic in a punctured neighborhood of 0. Define the *residue at infinity* to be

$$\mathrm{res}_\infty f := -\mathrm{res}_0 \frac{1}{w^2} f\left(\frac{1}{w}\right).$$

Let f be holomorphic on $\mathbb{C} - \{z_1, ..., z_n\}$ where $\{z_k\}$ are isolated singularities. Then,

$$\mathrm{res}_\infty f + \sum_{k=1}^{n} \mathrm{res}_{z_k} f = 0.$$

Essential singularities are a lot harder to deal with. An example of a function with an essential singularity is $f(z) := e^{\frac{1}{z}}$ at 0.

Theorem 37 (Casorati-Weierstrass): Let f holomorphic on a punctured disc $D \setminus \{z_0\}$ with an essentially singularity at z_0. Then $f(D \setminus \{z_0\})$ is dense in $\mathbb{C}$.

Proof. Argue by contradiction. ■

Definition 24 (Meromorphic Function, Singularities at Infinity): Let $f : \Omega \to \mathbb{C}$. We say f is *meromorphic* if there exists a sequence $\{z_n : n \geq 0\}$ of points in Ω such that

- $\{z_n\}$ has no limit points in Ω;
- f holomorphic on $\Omega \setminus \cup_n \{z_n\}$; and
- f has poles at each $\{z_n\}$.

Define $F(z) := f\left(\frac{1}{z}\right)$. We say f has a *pole/removable/essential singularity at infinity*/is *holomorphic at infinity* if F has a pole/removable/essential singularity/is holomorphic at 0. If f is a meromorphic function that is either holomorphic or has a pole at infinity is said to be *meromorphic in the extended complex plane*.

Proposition 12: The only meromorphic functions in the extended comlex plane are the rational functions.

Theorem 38 (Argument Principle): Let f be a meromorphic function on the interior of a simple closed curve $C \subset \mathbb{C}$, such that f has no roots nor poles on C. Then,

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,\mathrm{d}z = \#\{\text{zeroes of } f \text{ inside } C\} - \#\{\text{poles of } f \text{ inside } C\},$$

the right-hand side counted with multiplicity.

Proof. If z_0 a zero of multiplicity n of f, then it will be a pole of $\frac{f'}{f}$ with residue n, and if z_0 a pole of order n of f, then it will be a pole of $\frac{f'}{f}$ with residue $-n$. The result then follows by the residue theorem. ■

Remark 20: Sometimes the quantity $\frac{f'}{f}$ is called the *logarithmic derivative* of f, agreeing with the idea that "$\frac{\mathrm{d}}{\mathrm{d}z}\log(f) = \frac{f'}{f}$".

Corollary 4: Let f as above, C as above, and let $\arg(f)$ be the argument of f. Then,

$$\#\{\text{zeroes of } f \text{ inside } C\} - \#\{\text{poles of } f \text{ inside } C\} = 2\pi\Delta_C \arg(f),$$

where $\Delta_C \arg(f)$ the total change of the argument of f over C.

Proof. One notices $\frac{f'}{f} = \frac{\mathrm{d}}{\mathrm{d}z}\log(f(z))$. Split the log into ln of norm plus argument, then apply fundamental theorem of calculus. ■

Theorem 39 (Rouché's): Let f, g be holomorphic functions on Ω with a curve $C \subset \Omega$ such that

- $|f| > |g|$ on C,
- both f, g never vanish on C.

Then, f and $f + g$ have the same number of zeros in the interior of C.

Proof. Define $f_t = f + tg$ for $t \in [0,1]$. The conditions on f, g show that this function has no zeroes on C for any t, so we can apply the argument principle. Defining $N(t) = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,\mathrm{d}z$, one sees that this is a continuous (in t) function that is integer valued, and is therefore constant. In particular, this implies $N(0) = N(1)$, where the form is the number of zeros of f and the latter of $f+g$, inside C. ■

Theorem 40 (Maximum Modulus Principle): Let $f : \Omega \to \mathbb{C}$ a nonconstant holomorphic function. Then $|f|$ cannot attain its maximum inside Ω. In particular, if Ω bounded, then

$$\max_{\overline{\Omega}}|f| = \max_{\partial\Omega}|f|,$$

and if $|f|$ attains an "interior" maximum, then f is constant.

Theorem 41 (Logarithms): Let Ω a simply connected domain containing 1 and not containing 0. Then there exists a function $F(z) = \log_\Omega(z)$, called a *branch* of the logarithm, such that

- F holomorphic on Ω,
- $e^{F(z)} = z$ on Ω, and
- $F(r) = \log(r) := \int_1^r \frac{1}{x}\,\mathrm{d}x$ whenever r is a real number near 1 in Ω.

Proof. The idea is to define F by

$$F(z) = \int_{\gamma_z} \frac{1}{w}\,\mathrm{d}w,$$

where γ_z any curve connecting 1 to z. By simple connectedness, this integral is independent of choice of curve, and one checks the necessary properties rather easily by verifying $F'(z) = \frac{1}{z}$. In particular, the last follows form the fact that if $r \approx 1$, one can take γ_z to lie entirely on the real line.
■

Theorem 42 (Function Logarithms): Let f a non-vanishing holomorphic function on a simply connected domain Ω. Then there exists a holomorphic function g on Ω such that $f = e^g$.

1.2.3 Harmonic Functions

Theorem 43: Let f holomorphic on a disc $D_R(z_0)$ with series expansion $f(z) = \sum_{n\geq 0} a_n(z-z_0)^n$. Then, for any $0 < r < R$ and $n \geq 0$,

$$a_n = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,\mathrm{d}\theta.$$

In particular, the case $n = 0$ yields the *mean value property* for holomorphic functions.

Proof. This follows directly by parametrizing the circle in the Cauchy integral formula for the derivatives of f. ■

Proposition 13 (Mean Value Property for Harmonic Functions): Let $u : \Omega \to \mathbb{R}$ be a harmonic function on a simply connected domain Ω. Then there exists a holomorphic function $f : \Omega \to \mathbb{R}$ such that $u = \mathrm{Re}(f)$, and so in particular,

$$u(z_0) = \left(\frac{1}{2\pi r}\right)\int_0^{2\pi} u\left(z_0 + re^{i\theta}\right)\mathrm{d}\theta,$$

for any $z_0 \in \Omega$ and $r > 0$ such that $D_r \subset \Omega$.

Proof. We need to find v such that $f = u + iv$ is a holomorphic function. Such a v is called a *harmonic conjugate* of u. We can define f to be the antiderivative of $2\frac{\partial u}{\partial z}$, which, since $\mathbb{D}$ simply connected, exists. This would imply $f'(z) = u_x + iu_y$, which should hold. Moreover, $\Re(f) = u$ up to a constant (check!). The mean value property for harmonic functions then holds as a consequence of the previous theorem (with $n = 0$) and taking real parts of both sides of the identity. ■

Proposition 14 (Other Properties of Harmonic Functions): Let u be harmonic on Ω. Then:

- u cannot achieve an interior min/max, unless it is constant, and if Ω bounded, u obtains its absolute min/max on $\partial\Omega$
- if $\Omega = \mathbb{R}^2$ and u bounded, then u is constant
- u is real-analytic, and in particular C^∞

1.2.4 Conformal Mappings

Definition 25 (Conformal Map): A function $f : U \to V$, where $U, V \subset \mathbb{C}$ open, is said to be a *conformal mapping* if it is bijective and holomorphic. If U, V are given and there exists a conformal map $f : U \to V$, we say U, V are *conformal*.

Proposition 15: Let $f : U \to V$ be holomorphic and injective. Then:

- $f'(z) \neq 0$ for every $z \in U$
- f is invertible when restricted to its image, and its inverse is holomorphic

In particular, if f conformal, then it's inverse $f^{-1} : V \to U$ is automatically holomorphic, and its derivative never vanishes.

Proposition 16 (Conformal map from $\mathbb{D}$ to $\mathbb{H}$): Let $\mathbb{D}$ be the open unit disc in $\mathbb{C}$ and $\mathbb{H} := \{z \in \mathbb{C} \mid \operatorname{Im}(z) > 0\}$ be the open upper half-plane. Then $\mathbb{D}$ and $\mathbb{H}$ are conformal, under the mapping

$$F : \mathbb{H} \to \mathbb{D}, \quad F(z) := \frac{i - z}{i + z},$$

with inverse

$$G : \mathbb{D} \to \mathbb{H}, \quad G(w) := i\frac{1 - w}{1 + w}.$$

Lemma 1 (Schwarz's Lemma): Let $f : \mathbb{D} \to \mathbb{D}$ be a holomorphic function with $f(0) = 0$. Then the following hold:

- $|f(z)| \leq |z|$, and if there exists a $z_0 \neq 0$ such that $|f(z_0)| = |z_0|$, then $f(z) = cz$ where $|c| = 1$ (i.e. f a rotation)
- $|f'(0)| \leq 1$, and if equality holds, then $f(z) = cz$ where $|c| = 1$.

Proof. Let $g(z) := \frac{f(z)}{z}$. Since $f(0) = 0$, g has a removable singularity at the origin and therefore is holomoprhic. Let $z \in \mathbb{D}$, assuming $|z| = r < 1$. Since f maps to $\mathbb{D}$ and thus $|f| \leq 1$, we have that $|g(z)| \leq \frac{1}{r}$. By the maximum modulus principle, this bound must hold for all $z \in \mathbb{D}_r$ i.e. for any $|z| < r$. Letting $r \to 1$ gives that $|g| \leq 1$ which gives the proof. Again by the maximum modulus principle, we see that if $|g(z_0)| = 1$ for z_0 in $\mathbb{D}$ (i.e. an "interior max"), it must be that $|g|$ constant equal to 1, which gives the result. For the second point, note that $g(z) \to f'(0)$ as $z \to 0$, but also converges to $g(0)$ by holomorphicity. We know $|g| \leq 1$, so it follows that $|f'(0)| \leq 1$. Finally if $|f'(0)| = |g(0)| = 1$, from the same line of reasoning as above we have that f a rotation. ■

Theorem 44 (Automorphisms of the Unit Disc): For $\Omega \subset \mathbb{C}$, define $\operatorname{Aut}(\Omega) := \{f : \Omega \to \Omega \mid f \text{ a conformal map}\}$. This is naturally a group under composition.

If $f \in \operatorname{Aut}(\mathbb{D})$, then there exists a $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta}\psi_\alpha(z),$$

where $\psi_\alpha(z) := \frac{\alpha - z}{1 - \overline{\alpha} z}$ is a *Blaschke factor*.

Proof. Let $\alpha \in \mathbb{D}$ be such that $f(\alpha) = 0$, which exists and is unique. Let $g = f \circ \psi_\alpha$. One checks that $g(0) = 0$, so by Schwarz, $|g(z)| \leq |z|$. g is also invertible, and $g^{-1}(0) = 0$ so we also have $|g^{-1}(w)| \leq |w|$. This implies, taking $w = g(z)$, that $|z| \leq |g(z)| \leq |z|$ i.e. $|g(z)| = |z|$. This implies $g(z) = e^{i\theta} \cdot z$ for some $\theta \in \mathbb{R}$, which gives the proof upon composing both sides by ψ_α (which is its own inverse). ■

Corollary 5 (Automorphisms of the Upper Half Plane): Every automorphism in $\operatorname{Aut}(\mathbb{H})$ is a *linear fractional transformation* of the form

$$f_M(z) = \frac{az + b}{cz + d},$$

for some $M \in \mathrm{SL}_2(\mathbb{R}) = \{M \in \mathbb{R}^{2\times 2} : \det(M) = 1\}$. In fact,

$$\operatorname{Aut}(\mathbb{H}) \cong \mathrm{PSL}_2(\mathbb{R}) := \mathrm{SL}_2(\mathbb{R})/\{I, -I\}$$

Proof. Recall the conformal function $F : \mathbb{H} \to \mathbb{D}$. One checks that $\Gamma : \operatorname{Aut}(\mathbb{D}) \to \operatorname{Aut}(\mathbb{H})$ given by $\Gamma(\varphi) := F^{-1} \circ \varphi \circ F$ is a group isomorphism. Thus to characterize automorphisms of $\mathbb{H}$ it suffices to compute the conjugation by F of automorphisms of $\mathbb{D}$, which yields the result. ■

Theorem 45 (Riemann-Mapping Theorem): Let $\Omega \subset \mathbb{C}$ be a simply connected open domain that is not all of $\mathbb{C}$. Given $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that

$$F(z_0) = 0, \quad F'(z_0) > 0.$$

In particular, every such domain is conformal to the open unit disc.

Proof. The proof is long. The idea is as follows:

(i) Show Ω conformal to an open subset of $\mathbb{D}$ containing the origin. One does this by basically 'shrinking' Ω via a logarithmic-type transformation.

(ii) Assuming from 1. that $0 \in \Omega \subset \mathbb{D}$, one considers the family of functions

$$\mathcal{F} := \{f : \Omega \to \mathbb{D}, f \text{ holomorphic, injective, and } f(0) = 0\}.$$

One sees that $\mathcal{F}$ is a nonempty, uniformly bounded family of holomorphic functions. By the Cauchy inequalities, $s := \sup_{f \in \mathcal{F}} |f'(0)|$ exists (and is finite). Let $\{f_n\} \subset \mathcal{F}$ such that $|f_n'(0)| \to s$. By combining Cauchy's integral representation theorem, one obtains equicontinuity of $\mathcal{F}$, which allows us to use an Arzela-Ascoli-type argument to argue that $\{f_n\}$ actually converges uniformly (on compact sets) to some holomorphic function f. We see that f is injective (by an "argument principle"-type argument, using properties of f).

(iii) We claim f in step 2. is conformal. We showed holomorphicity and injectivity, so we need to show surjectivitiy. Supposing otherwise, the idea is to construct another function that lives in $\mathcal{F}$ but with strictly greater derivative norm at 0, contradiciting the maximality. Then by precomposing f with a rotation, one automatically gets the $f'(0) > 0$.

■

Corollary 6: Any two simply connected proper open subsets of $\mathbb{C}$ are conformal.

Remark 21: Remark that there is no hope that there exists a conformal map $f : \mathbb{C} \to \mathbb{D}$ for such a map would necessarily be entire and bounded and thus constant by Liouville's, and in particular not injective.

1.2.5 Some Fourier Transform

Definition 26 (Fourier Transform): The *Fourier transform* of a function $f : \mathbb{C} \to \mathbb{C}$ is defined by, where the integrals make sense and converge,

$$\hat{f}(\xi) := \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} \, dx, \quad \xi \in \mathbb{R}.$$

For $a > 0$, define the class of functions

$$\mathcal{F}_a := \left\{ f : \begin{array}{c} f \text{ holomorphic on } S_a := \{z \in \mathbb{C} \mid |\mathrm{Im}(z)| \leq a\} \\ \exists\, A > 0 \text{ such that } |f(z)| \leq \frac{A}{1+\Re(z)^2} \text{ for all } z \in S_a \end{array} \right\}.$$

Define finally $\mathcal{F} := \bigcup_{a>0} \mathcal{F}_a$.

Proposition 17: Let $f \in \mathcal{F}_a$ for some $a > 0$. Then $|\hat{f}(\xi)| \leq B e^{-2\pi b\, |\xi|}$ for all $\xi \in \mathbb{R}$ and $0 \leq b < a$; when such a bound on a function exists, we say it is of *exponential type*.

Proof. The idea is to "shift" the contour to the line from $-\infty - ib$ to $\infty - ib$. ■

Theorem 46 (Fourier Inversion): For $f \in \mathcal{F}$, then the *Fourier inversion formula* holds, that is, for all $x \in \mathbb{R}$,

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} \, d\xi.$$

Theorem 47 (Poisson Summation Formula): If $f \in \mathcal{F}$,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n).$$

2 Algebra

2.1 Linear

2.1.1 Elementary

2.1.2 Vector Spaces

2.1.3 Diagonalization and Related

2.2 Groups

2.2.1 Fundamentals

Definition 27 (Groups, Action): A group G is a set equipped with a binary operation $\cdot$ satisfying the axioms:

- $f \cdot (g \cdot h) = (f \cdot g) \cdot h,\ \forall\, f, g, h \in G$
- $\exists\, 1 \in G$ s.t. $g \cdot 1 = 1 \cdot g = g\, \forall\, g \in G$
- $\forall\, g \in G, \exists\, g^{-1} \in G$ s.t. $g \cdot g^{-1} = g^{-1} \cdot g = 1$

G is said to be abelian/commutative if $gh = hg$ for every $g, h \in G$.

A group action on a set X is a function $\varphi : G \times X \to X$ satisfying:

- $\varphi(h, \varphi(g, x)) = \varphi(h \cdot g, x), \forall\, g, h \in G, x \in X$
- $\varphi(1, x) = x\, \forall\, x \in X$

When clear from context, we write $g \cdot x = \varphi(g, x)$. When $\varphi(G, X) := \{\varphi(g, x) : g \in G, x \in X\}$ is equal to X, we say the action of G is *transitive*. A set X equipped with a group action of G is sometimes called a *G-set*.

Definition 28 (Orbit, Stabilizer): Let G act on a set X. Define, for $x \in X$,

$$\begin{aligned} \mathcal{O}_x &= \mathrm{Orb}_G(x) := \text{orbit of } x \text{ under } G = \{g \cdot x : g \in G\} \subset X, \\ G_x &= \mathrm{Stab}_G(x) := \text{stabilizer of } x \text{ under } G = \{g \in G : g \cdot x = x\} \subset G. \end{aligned}$$

Remark 22: The orbit is "everywhere x can go" (under G), and the stablizer is what "fixes" x (under G).

Definition 29: A *subgroup* H of a group G is a subset of G which is still a group when endowed with the operation from G (so in particular it is closed under the operation). One sometimes write $H < G$, or $H \leq G$ if H possibly equal to G.

A *(left) coset* of H in G is a set of the form

$$aH = \{ah : h \in H\},$$

where a some element of G. One denotes $G/H = \{\text{set of cosets of } H\}$ Equivalently, there is a natural action of H acting on G as a set (by right-multiplication), given by $h \star g := g \cdot h$; thus cosets of H are just orbits under this action.

Definition 30 (Homomorphism): A *group homorphism* $\varphi : G_1 \to G_2$ is a function such that

$$\varphi(gh) = \varphi(g)\varphi(h)$$

for all $g, h \in G_1$. When φ a bijection we call it a group isomorphism.

Theorem 48 (Lagrange): Let $H \subset G$ a finite subgroup of a finite group. Then all cosets of H have the same cardinality, and in particular are disjoint. In particular, we have

$$|G| = |G/H| \cdot |H|,$$

and so $|H|$ divides $|G|$.

Proof. The disjointness follows from the "orbit of an action" characterization (namely, orbits are either disjoint or equal). For the equality of cardinality, fix $a \notin H$ and define the map $f : H \to aH, h \mapsto ah$. This is an injection, since $f(h_1) = f(h_2) \Rightarrow ah_1 = ah_2 \Rightarrow h_1 = h_2$ by multiplying both sides by a^{-1}. This is also a surjection since given $ah \in aH$, one simply maps h by f. This shows $|aH| = |H|$ and since a was arbitrary, completes the proof. The "in particular" follows from the partitioning result. ■

Proposition 18: Let X a transitive G-set. Then there exists a subgroup H of G such that X is isomorphic (as a G-set) to $X \cong G/H$, equipped with the left-action

$$(g, aH) \in G \times G/H \mapsto g \cdot aH := (ga)H \in G/H.$$

Thus, $|X|$ divides $|G|$ and the *orbit-stabilizer formula* holds:

$$|G| = |X| \cdot |G_x|, \quad \forall\, x \in X.$$

Remark 23: When we say "X_1, X_2 isomorphic as G-sets", we mean a bijection which respects the respect group actions, i.e. $f : X_1 \to X_2$, $f(g \cdot_1 x) = g \cdot_2 f(x)$.

Remark 24: If X not transitive, we can reword this proof by taking any orbit $\mathcal{O}_x$ in X, in which case the formula reads $|G| = |\mathcal{O}_x| \cdot |G_x|$.

Proof. Fix $x \in X$ and let $H := G_x$, and let $f : G/H \to X$ be given by $f(aH) := a \cdot x$. ■

Definition 31 (Normal subgroup, quotients): Given a group G, a subgroup $H \subset G$ is said to be *normal* if it is closed under conjugation by G, that is,

$$ghg^{-1} \in H, \quad \forall\, g \in G, h \in H.$$

Sometimes people write $H \triangleleft G$ for "H is a normal subgroup of G" (*sometimes with a bar underneath if H possible equal to G*).

The *quotient group* is the group G/H of cosets of G with respect to H equipped with left-multiplication (*one checks that this is indeed a well-defined group when H normal in G).*

G is said to be *solvable* if there exists a sequence of subgroups

$$\{1\} = G_0 \subset G_1 \subset \cdots \subset G_{n-1} \subset G_n \subset G$$

such that G_{i-1} a normal subgroup of G_i and G_i/G_{i-1} abelian, for $i = 1, ..., n$.

Proposition 19: Let $\varphi : G \to H$ a group homomorphism. Then $N := \ker(\varphi)$ a normal subgroup of G.

Moreover, φ induces an injective homomorphism $\tilde{\varphi} : G/N \to H$ defined by $\tilde{\varphi}(aN) := \varphi(a)$. In particular, $\mathrm{im}(\varphi)$ is isomorphic to G/N.

Proof. If $h \in N, g \in G$, then $\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g)^{-1} = \varphi(g) \cdot \varphi(g)^{-1} = 1$ so $ghg^{-1} \in N$.

We need to show $\tilde{\varphi}$ well-defined. If $aN = a'N$, then there exists $h \in N$ such that $a = a'h$. So

$$\tilde{\varphi}(aN) = \varphi(a) = \varphi(a'h) = \varphi(a')\varphi(h) = \tilde{\varphi}(a'N) \cdot 1 = \tilde{\varphi}(a'N),$$

so the function is well-defined. It is injective since

$$\tilde{\varphi}(aN) = 1 \Leftrightarrow \varphi(a) = 1 \Leftrightarrow a \in N \Leftrightarrow aN = N,$$

but $N = 1_{G/N}$, so φ injective. ■

2.2.2 Sylow

Theorem 49 (Sylow's Theorem): Suppose $\#G = m \cdot p^t$, where $p \nmid m$ and p prime.

(i) (*Sylow 1*) G has a subgroup of cardinality p^t (called a *Sylow-p subgroup* of G).

(ii) (*Sylow 2*) Suppose H_1, H_2 are Sylow-p subgroups of G, then there exists a $g \in G$ such that $gH_1g^{-1} = H_2$.

(iii) (*Sylow 3*) If N_p the number of distinct Sylow-p subgroups of G, then:
   - (i) $N_p \mid m$
   - (ii) $N_p \underset{p}{\equiv} 1$

Proof. ■

2.2.3 Some Particular Groups

Definition 32 (Generator of a group): We say a group G is *generated* by elements $g_1, ..., g_n$ if

$$G = \langle g_1, ..., g_n \rangle := \{\text{all possible products of powers of } g_1, ..., g_n\}.$$

Definition 33 (Cyclic Groups, Unit Groups): For $n \geq 1$ an integer, the *cyclic group of order n* is defined as $\mathbb{Z}/n\mathbb{Z} = \{0, 1, ..., n-1\}$ equipped with addition and with entries read mod n.

For $n \geq 1$ an integer, the *unit group of order n* is defined as $(\mathbb{Z}/n\mathbb{Z})^\times$ equipped with multiplication, where we restrict $\mathbb{Z}/n\mathbb{Z}$ to the invertible elements (mod n).

Proposition 20: Let p prime and G a group of order p. Then $G = \mathbb{Z}/p\mathbb{Z}$. In addition, $\#(\mathbb{Z}/p\mathbb{Z})^\times = p - 1$.

Proof. Let $a \in G$ not equal to the identity. Then the subgroup H generated by a (i.e. $\{a, a+a, a+a+a, ...\}$) is a subgroup of G isomorphic to $\mathbb{Z}/n\mathbb{Z}$ where $n = \#H$. Being a subgroup, $n | \#G$, and since $n > 1$ and $\#G$ prime, $n = p$ and thus $H = G \cong \mathbb{Z}/p\mathbb{Z}$.

Let $a \in (\mathbb{Z}/p\mathbb{Z}) \setminus \{0\}$. Then $a \nmid p$ so $\gcd(a, p) = 1$. Thus there exists a $b, c \in \mathbb{Z}$ such that $ab + cp = 1$ i.e. $ab \underset{p}{\equiv} 1$. Rewriting b mod p so that $b \in (\mathbb{Z}/p\mathbb{Z}) \setminus \{0\}$ completes the proof of the second part, since then we've shown $((\mathbb{Z}/p\mathbb{Z})^\times$ equals (as a set) $(\mathbb{Z}/p\mathbb{Z}) \setminus \{0\})$. ■

Definition 34 (Permutation, Symmetric, Alternatic Groups): Let X a finite set of n elements. The *symmetric group* of X is the set of bijections of X, i.e.

$$S_X := \{f : X \to X \mid f \text{ a bijection}\}.$$

When $X = \{1, ..., n\}$, we often write $S_n \equiv S_X$. We call $f \in S_X$ a *permutation* of X.

The *sign*, or sgn, of a permutation $f \in S_n$ is defined to be

$$\operatorname{sgn}(f) := (-1)^{N(f)}, \quad N(f) := \#\{(x, y) \in X \times X \mid x < y \text{ and } f(x) > f(y)\},$$

i.e. the number of "inversions" that f contains. We sometimes say f is even/odd if $\operatorname{sgn}(f) = +1/-1$.

The *alternating group* on n elements is the subgroup

$$A_n := \{f \in S_n \mid \operatorname{sgn}(f) = 1\}.$$

Proposition 21:

- $\#S_n = n!$, $\#A_n = \frac{n!}{2}$
- a *transposition* of $\{1, ..., n\}$ is a permutation that only interchanges two elements and fixes the rest. Any permutation can be written as a composition of transpositions. Moreover, the sign of a permutation is equal to the parity of (any) number of transpositions needed to write the permutation

Proof. A permutation, being a bijection of $\{1, ..., n\}$ has n choices of where to send 1, $n-1$ choices of where to send 2, etc, for $n \cdot (n-1) \cdots 1 = n!$ total choices. So, $\#S_n = n!$. The size of $\#A_n$ follows pretty clearly from the definition of sgn. Alternatively, one notes that $\operatorname{sgn} : S_n \to \{-1, 1\}$ a surjective group homomorphism (with the right-hand side equipped with multiplication). Then $A_n = \ker(\operatorname{sgn})$, and thus $(\#S_n)/(\#A_n) = 2$. ■

Definition 35 (Dihedral Groups): For $n \geq 1$, the *order $2n$ dihedral group* (on n vertices) is defined as the gorup generated by

$$D_{2n} := \langle \sigma, \tau \rangle, \quad \operatorname{ord}(\sigma) = 2, \operatorname{ord}(\tau) = n, \sigma\tau\sigma^{-1} = \tau^{-1}.$$

Proposition 22: $\#D_{2n} = 2n$.

Proposition 23 (Finitely Generated Abelian Groups): An abelian group G is said to be *finitely generated* if it has a finite generating set. Any finitely generated abelian group G is isomorphic to

$$\mathbb{Z}^m \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_m\mathbb{Z}.$$

2.3 Rings

Definition 36 (Rings, Ideals): A *ring* is a set R equipped with two binary operators $+$ and $\times$ such that

- $(R, +)$ an abelian group
- $(R, \times)$ a monoid (has identity and is associative)
- $a \times (b + c) = a \times b + a \times c$
- $(b + c) \times a = b \times a + c \times a$

A commutative ring is a ring where $\times$ is commutative.

An *ideal* I of a ring R is a subset $I \subset R$ such that I is an additive subgroup of R and is closed under multiplication by R. We sometimes write $I \triangleleft R$.

Remark 25 : Some conventions do not require rings to have multiplicative identities, and call those with such elements "rings with identity".

Definition 37 (Types of Ideals): Let $I \triangleleft R$. We say I is:

- *maximal* if for all ideals $J \triangleleft R$, if $I \subsetneq J$ then $J = R$
- *prime* if $a \cdot b \in I \Rightarrow a$ or $b \in I$
- *principal* if I of the form $aR = (a) = \{ar : r \in R\}$ for some $a \in R$

Definition 38 (Types of Rings): Let R be a nonzero ring. A *left (right) zero divisor* of R is an element $a \in R$ such that there exists a nonzero $b \in R$ such that $ab = 0$ ($ba = 0$). We say R is:

- an *integral domain* if it is commutative and has no nonzero zero divisors
- a *principal ideal ring* if every ideal in R is principal
- a *Euclidean domain* if it is an integral domain upon which there exists a function $f : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ such that for all $a \in R, b \in R \setminus \{0\}$ there exists q and r in R such that $a = bq + r$ and either $r = 0$ or $f(r) < f(b)$
- a *principal ideal domain (PID)* if it is an integral domain and a principal ideal ring

Remark 26 : The f function, called a *Euclidean function*, is essentially a generalization of "size", which is not necessarily an ordering but reflects some of its properties.

Proposition 24: Euclidean domain $\Rightarrow$ PID. The integers are a Euclidean domain. If $\mathbb{F}$ a field, then $\mathbb{F}[x] = \{p(x) : p \text{ a polynomial with coefficients in } \mathbb{F}\}$ is a Euclidean domain.

PROOF. Let R a Euclidean domain and $I \subset R$ an ideal. Then, $S := \{f(r) \mid r \in I\}$ is a subset of the natural numbers and thus must have a least element by the Well-Ordering Principle. Let $m \in I$ be such that $f(m) = \min S$. We claim $I = (m)$. The direction $(m) \subset I$ is clear since $m \in I$. Now let $a \in I$ be any other element. By the Euclidean property, $a = mq + r$ for some $q, r \in R$. If $r = 0$ we're done, so assume $r \neq 0$ so then $f(r) < f(m)$. This implies r cannot be in I, else we'd contradict the minimality of m. But $a - mq \in I$ (since $a, m \in I$, and $q \in R$ implies $mq \in I$), but this equals r, so we have a contradiction. Thus $r = 0$, and so $a = mq \in (m)$, hence $I = (a)$.

The integers are Euclidean by taking $f(r) = r$ and by employing the Euclidean algorithm, and $\mathbb{F}[x]$ is by taking $f(p) = \deg(p)$ and employing the Euclidean algorithm (for polynomials). ∎

Definition 39 (Homomorphism): A ring homomorphism $\varphi : R \to S$ is a map that is a group homomorphism on $(R, +)$, is multiplicative (i.e. $\varphi(ab) = \varphi(a)\varphi(b)$), and $\varphi(1) = 1$.

Proposition 25: Given a ring homomorphism $\varphi : R \to S$, $\ker(\varphi)$ an ideal of R. Conversely, if $I \subset R$ is an ideal, then there exists a ring S and a surjective homomorphism $\varphi : R \to S$ such that $I = \ker(\varphi)$.

If $\varphi : R \to S$ is a surjective ring homomorphism, then $R/\ker(\varphi)$ is isomorphic to S.

PROOF. For the first, we know from the group homomorphism property that $\ker(\varphi)$ closed under addition, and if $r \in \ker(\varphi)$ and $a \in R$, then $\varphi(ar) = \varphi(a)\varphi(r) = \varphi(a) \cdot 0 = 0$ so $ar \in \ker(\varphi)$. For the second, we can define the *quotient ring* $S := R/I = \{a + I : a \in R\}$, with addition defined as in a quotient group and multiplication defined "component wise" (one checks this is well-defined and defines a ring). Then define $\varphi : R \to S$ by $\varphi(r) = r + I$.

The last point is the *first isomorphism theorem* (for rings) and the proof is identical as the analog for groups. ∎

Proposition 26: Let R a ring.

- I is a prime ideal iff R/I has no nonzero zero divisors
- I is a maximal ideal iff R/I is a field

Proof. Assume I prime and that $(a+I)(b+I) = 0+I$. This implies $ab \in I$, so either a or b in I by primality. Thus either $a+I$ or $b+I = I$, and in either case neither can be a nonzero zero divisor. The converse direction is identical.

Assume I maximal. We need to show R/I has inverses. Let $a+I \in R/I$ nonzero (i.e. $a \notin I$). Consider the ideal $J := Ra + I = \{ra + b : r \in R, b \in I\}$. This is an ideal which contains I as a subset. By maximality, $J = I$ or $J = R$, but the first is not possible since this would imply $a \in I$. So $J = R$, and thus given $1 \in R$, there exists $r \in R$ and $b \in I$ such that $ar + b = 1$. Thus,

$$(a+I)(r+I) = ar + I = ar + b + I = 1 + I,$$

i.e. $r + I = (a+I)^{-1}$. Conversely, assume $J \supsetneq I$. Let $a \in J \setminus I$, so $a + I \neq 0 \in R/I$ is invertible, i.e. there is a $b \in R$ such that $(a+I)(b+I) = 1+I$. This implies there is an $r \in I$ such that $ab + r = 1$. Since also $r \in J$, this implies $1 \in J$, and thus $J = R$ and so I maximal. ■

2.4 Fields & Galois Theory

Definition 40 (Fields, Field Extensions): A *field* is a commutative ring F with identity in which every nonzero element has a multiplicative inverse.

A *field extension* E of a field F is a field which contains F as a subfield; we write E/F. We can canonically view E as an F vector space (by forgetting the multiplicative structure of elements); we write $[E : F] := \dim_F(E)$ for the *degree* of E over F. We say E/F a *finite extension* if this number is finite.

Proposition 27 (Multiplicativity of Degrees): Let K/E and E/F be finite extensions. Then K/F also a finite extension, and

$$[K : F] = [K : E] \cdot [E : F].$$

Proof. If $\{e_1, ..., e_n\}$ a basis for E/F and $\{k_1, ..., k_m\}$ a basis for K/E, one checks that $\{e_i \cdot k_j : 1 \le i \le n, 1 \le j \le m\}$ a basis for K/F. ■

Definition 41 (Algebraic, Transcendental): Let E/F. We say $\alpha \in E$ is *algebraic* if it is the root of some polynomial $f(x) \in F[x]$. We say α is *transcendental* otherwise.

Proposition 28: If E/F finite, every $\alpha \in E$ is algebraic. Moreover, there exists a polynomial of degree at most $[E : F]$ that α satisfies.

Proof. Let $\alpha \in E$ and put $n := [E : F]$. Then $\{1, \alpha, \alpha^2, ..., \alpha^n\}$ must be a linearly independent set, and thus there exist $f_0, ..., f_n \in F$ such that

$$f_0 + f_1\alpha + \cdots + f_n\alpha^n = 0.$$

In particular, we see that this implies $f(x) := f_n x^n + \cdots + f_1 x + f_0$ is a polynomial in $F[x]$ with α as a root. ■

Definition 42 (Splitting Fields): We say E/F a *splitting field* of a polynomial $f(x) \in F[x]$ if

(i) $f(x)$ factors into linear factors in $E[x]$, i.e. there exists $r_1, ..., r_n \in E$ such that

$$f(x) = (x - r_1)\cdots(x - r_n),$$

and

(ii) E is *generated* by $r_1, ..., r_n$.

Remark that $n \le [E : F] \le n!$.

Remark 27: We say a field extension E/F is *generated by a set* S if it is the smallest field containing F as a subfield and the elements in S.

Given an element $\alpha \in E/F$, a polynomial $f \in F[x]$ is called a *minimal polynomial* for f if $f(\alpha) = 0$ and f of minimal degree.

Theorem 50 (Eiseinstein's Criterion): Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$. Assume there exists a prime p such that:

(i) $p \mid a_i$ for $0 \le i < n$

(ii) $p \nmid a_n$

(iii) $p^2 \mid a_n$

Then $f(x)$ is irreducible over $\mathbb{Q}$.

Theorem 51 (Rational Root Theorem): Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ with $a_n \ne 0$. Then if $x = \frac{p}{q}$ (reduced) a rational root of f, then $p | a_0$ and $q | a_n$.

Theorem 52 (Gauss's Lemma):

Definition 43 (Automorphisms of a Field Extension): Let E/F, and define the group

$$\mathrm{Aut}(E/F) := \{\sigma : E \to E \mid \sigma \text{ is } F-\text{linear, multiplicative, and } \sigma|_F \equiv \mathrm{id}\}.$$

Remark 28: $\sigma \in \mathrm{Aut}\,(E/F)$ preserves the field structure on E and leaves F invariant.

Proposition 29 (Properties of $\mathrm{Aut}(E/F)$): Let E/F be a finite extension. Then the following hold:

- $\mathrm{Aut}(E/F)$ acts on E with finite orbits (that is, $\#\mathrm{Orb}_{\mathrm{Aut}(E/F)}(\alpha) < \infty$ for every $\alpha \in E$)
- $\#\mathrm{Aut}(E/F) < \infty$; in fact,
- $\#\mathrm{Aut}(E/F) \leq [E : F]$.

If $\#\mathrm{Aut}(E/F) = [E : F]$, we say that E/F a *Galois extension*, and write $\mathrm{Gal}(E/F) = \mathrm{Aut}(E/F)$.

Proof. Let $\alpha \in E/F$ satisfy $f(x) = a_n x^n + \cdots + a_0$ (exists by finiteness). Let $\sigma \in \mathrm{Aut}(E/F)$. Then notice, using linearity and multiplicativity of σ,

$$\begin{aligned} f(\sigma(\alpha)) &= a_n \sigma^n(\alpha) + \cdots + a_1 \sigma(\alpha) + a_0 \\ &= a_n \sigma(\alpha^n) + \cdots + a_1 \sigma(\alpha) + a_0 \\ &= \sigma(a_n \alpha^n + \cdots + a_1 \alpha + a_0) \\ &= \sigma(f(\alpha)) = \sigma(0) = 0. \end{aligned}$$

Thus, $\sigma(\alpha) \in \{\text{roots of } f\}$ hence $\mathrm{Orb}\ (\alpha) \subset \{\text{roots of } f\}$, proving the finiteness since f has only finitely many roots. By a previous proposition, this moreover shows $\#\mathrm{Orb}(\alpha) \leq [E : F]$.

Let $e_1, ..., e_n$ a basis for E/F and $\sigma \in \mathrm{Aut}(E/F)$. By linearity, σ uniquely determined by the n-tuple $S_\sigma := (\sigma(e_1), ..., \sigma(e_n))$. We see that $S_\sigma \in \mathrm{Orb}(e_1) \times \cdots \times \mathrm{Orb}(e_n)$. The set on the right-hand side is finite (with size at most n^n by the previous proof) so $\#\mathrm{Aut}(E/F) < \infty$.

■

Proposition 30: Let E/F a finite Galois extension with Galois group G. Then, $E^G := \{\alpha \in E : g\alpha = \alpha \,\forall\, g \in G\}$ is a subfield of E containing F and moreover $E^G = F$.

Proof. Checking subfield is clear. Consider the group $\mathrm{Aut}(E/E^G)$. This contains G as a subgroup, so

$$[E : F] = \#G \leq \#\mathrm{Aut}(E/E^G) \leq [E : E^G].$$

Also $[E : E^G]$ divides $[E : F]$, which is then only possible if $[E : E^G] = [E : F]$ so it must $E^G = F$. ■

Proposition 31: Let E/F Galois. If f irreducible over F and has a root in E, then f splits completely into linear factors in E.

Proof. Let r be a root. Let $\{r_1, ..., r_n\}$ be the orbit of r under G. Let

$$g(x) = (x - r_1) \cdots (x - r_n) = x - \sigma_1 x^{n-1} + \cdots + (-1)^n \sigma_n,$$

where σ_i's are called *elementary symmetric functions*. Note that each σ_i invariant under permutation by G, and in particular, $\sigma_i \in E^G$. It follows that each $\sigma_i \in F$, by Galois and so $g(x) \in F[x]$. Recalling that f irreducible over F, it must be the minimal polynomial of r and thus $f | g$. Since g splits into linear factors (in $E[x]$), so must f. ■

2.4.1 Characterization of Finite Fields

Theorem 53: Let F be a finite field (i.e. as a set, F is finite). Then, $\#F = p^n$ for some prime p and an integer $n \geq 1$.

Conversely, given a prime p and an integer $n \geq 1$, there exists a unique (up to isomorphism) field of cardinality p^n.

Proof. For the first, since F a finite field, there must exist some minimal integer p such that $1 + \cdots + 1$ (p-times) equals 0. p must be prime, else this would imply F has zero divisors. In particular, this shows $\mathbb{F}_p = (\mathbb{Z}/p\mathbb{Z})$ is a subfield of F; we can view F as a vector field over $\mathbb{F}_p$, and letting $n = \dim_{\mathbb{F}_p}(F)$, it's clear that $\#F = p^n$.

For the second point, let $f(x) = x^{p^n} - x$ as a polynomial over $\mathbb{F}_p$ and let F be its splitting field. We claim $\#F = p^n$. Indeed, $f'(x) = -1$ in F and so $\gcd(f, f') = 1$ and so f has no repeated roots. Thus, $\#F \geq p^n$. Moreover, one remarks that the set of roots of f is itself a field over $\mathbb{F}_p$, thus $\#F \leq p^n$. Uniqueness follows from uniqueness of splitting fields. ■

2.4.2 Fundamental Theorem of Galois Theory

Lemma 2: If E/F Galois and $K \subset E$, then E/K Galois. If H a subgroup of the Galois group $\mathrm{Gal}(E/F)$, then E^H a subfield of G and $H = \mathrm{Gal}(E/E^H)$, and in particular $\#H = [E : E^H]$.

Remark 29: There is *no guarantee* that K/F is Galois in general.

Proposition 32 (Galois Correspondance): Let E/F a Galois extension. Then the map

$$\{K \text{ field} \mid F \subset K \subset E\} \to \{\text{subgroups of } \mathrm{Gal}(E/F)\}, \quad K \mapsto \mathrm{Gal}(E/K)$$

is a bijection with inverse

$$H \mapsto E^H.$$

Moreover, these maps are "order reversving" in the sense that

$$K_1 \subset K_2 \Rightarrow \mathrm{Gal}(E/K_1) \supset \mathrm{Gal}(E/K_2), \quad H_1 \subset H_2 \Rightarrow E^{H_1} \supset E^{H_2}.$$

Proposition 33: Let $F \subset K \subset E$ with E/F Galois. Then TFAE:

(i) $\sigma K = K$ for all $\sigma \in \mathrm{Gal}(E/F)$

(ii) K Galois over F

(iii) $\mathrm{Gal}(E/K)$ is a normal subgroup of $\mathrm{Gal}(E/F)$

Definition 44: An extension E/F is called *radical* if there exists an integer $n \geq 1$ and element $a \in F$ such that $E = F(a^{1/n})$. E/F a *tower of radical extensions* if there is a sequence of extensions $E = E_0 \subset E_1 \subset \cdots \subset E_n = F$ where each $E_i/(E_{i-1})$ a radical extension.

Theorem 54: If E/F a tower of radical extensions, then it is contained in Galois extension with solvable Galois group.

Theorem 55 (Fundamental Theorem of Galois): Assume F a field of characteristic 0. Suppose $f(x) \in F[x]$ is *solvable by radicals*, that is, a splitting field of f is contained in a radical extension. Then, $\mathrm{Gal}(f)$ is a solvable group.

3 Differential Equations

3.1 Scalar ODEs

3.1.1 First Order

3.1.2 Second Order, Constant Coefficients

3.1.3 Reduction of Order

3.1.4 Variation of Parameters

3.1.5 Sturm-Liouville Theory

3.1.6 Power Series Solutions

3.2 Linear Vector ODEs

3.2.1 Constant Coefficient Linear Systems

3.2.2 The Method of Undetermined Coefficients

3.3 Nonlinear Vector ODEs

3.3.1 Critical Points and Stability

3.4 General ODEs

3.4.1 Higher Order to Vector First Order

3.4.2 Laplace Transform

3.5 PDEs

3.5.1 Heat Equation

3.5.2 Wave Equation

3.5.3 Laplace's Equation

3.5.4 Separation of Variables

3.5.5 Laplace, Fourier Transforms Solutions to the Heat Equation

4 Miscellaneous Things to Remember

4.1 Trigonometric Identities

$$\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}, \quad \sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$$

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$
$$\cos(2\theta) = 1 - 2\sin^2(\theta)$$

Remark 30 : The easiest way to prove these is to use complex exponentials:

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$
$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$